

22 June 2026

Dr. Andrea Fuller
Editor-in-Chief
Conservation Physiology

Dear Dr. Fuller, Dr. Rodgers, and Reviewers,

We sincerely thank you all for your thoughtful and constructive evaluations of our manuscript titled “Xenobiotic Exposure May Confound Traditional Stress Biomarkers: Evaluating PFAS and Cadmium Effects on Largemouth Bass (*Micropterus nigricans*) Leukocyte Profiles” by Johnson *et al.* for consideration as a Research Article in *Conservation Physiology*. We appreciate the recognition that the topic is timely and relevant to conservation physiology, and we agree that the original version required substantial revision to improve methodological clarity, statistical transparency, ecological framing, and interpretation.

In response, we have extensively revised the manuscript. Major changes include: clarifying the experimental unit and repeated-measures structure; correcting and clarifying the neutrophil-to-lymphocyte ratio calculation; expanding methodological details on exposure preparation, water changes, salinity verification, feeding, and statistical modeling; adding sample-size information by treatment and sampling period; tempering statements that equated NLR directly with physiological stress; reframing the high-PFPeA treatment as a proof-of-concept toxicological exposure rather than eluding to an environmentally representative concentration; and expanding the Discussion to include plausible mechanistic pathways linking PFAS exposure, leukocyte dynamics, endocrine disruption, immune modulation, and altered stress physiology.

We have also added a more explicit paragraph on limitations to acknowledge the small sample size, the use of a single physiological endpoint, and the limited ecological realism of the high-PFPeA concentration. These revisions were made to ensure that the manuscript’s conclusions are appropriately cautious and consistent with the strength of the data.

In the three attached documents, we provide a (1) point-by-point response to each comment, (2) the original manuscript with all changes tracked, and (3) the updated manuscript.

Thank you again for your time and consideration. Please let me know if you need anything else from us at this time.

Sincerely,



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